

Causal-DFM: Diffusion-Based Causal-Invariant Dynamic Factor Model for Stock Return Prediction

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Abstract

Stock return prediction is highly sensitive to market regime shifts, such as bull-bear transitions, financial crises, and monetary policy cycles, where statistical relationships between observed features and returns can change abruptly. While recent deep learning-based factor models achieve strong performance in relatively stable periods, they often rely on correlations that are specific to historical market environments, causing learned patterns to break down when the market enters a different regime and leading to severe out-of-distribution (OOD) degradation. To address this challenge, we propose the Causal-Invariant Dynamic Factor Model (Causal-DFM), a causality-inspired factor-based framework that encourages compact and stable return-relevant representations while suppressing environment-dependent variations, thereby improving robustness under distribution shift. Concretely, Causal-DFM couples a conditional diffusion factor prior that captures regime-varying market dynamics with an Information-Bottleneck-regularized Transformer state encoder that distills a compact, return-predictive representation, suppressing spurious regime-specific signals while preserving the most informative content for prediction. The diffusion component provides a flexible generative mechanism for systematic market factors, while the Information Bottleneck regularization is implemented via an auxiliary return-prediction proxy that encourages the learned representation to be highly predictive yet capacity-controlled. We further present an information-theoretic analysis that motivates this objective and clarifies the conditions under which the learned representation can align with stable return mechanisms across environments. Empirical results on real-world stock datasets demonstrate that Causal-DFM consistently outperforms strong baselines, with particularly pronounced gains during regime transitions and crisis periods, highlighting its effectiveness in mitigating regime-induced OOD failures. The source code is available at https://github.com/bjtu-yzh/Causal-DFM.

CCS Concepts

• **Applied computing** → **Economics**; • **Information systems** → *Data stream mining*; Spatial-temporal systems; • **Computing methodologies** → Artificial intelligence.

Keywords

Stock Return Prediction, Causal Representation Learning, Distribution Shift, Diffusion Models, Dynamic Factor Model

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Resource Availability:
The source code of this paper has been made publicly available at https://github.com/bjtu-yzh/Causal-DFM.

1 Introduction

Stock return prediction is a core problem in quantitative finance. The Multi-Factor Model (MFM) attributes returns to a small set of common risk factors. Classical linear factor models, from the single-factor model [29] to the Fama-French three-factor [9] and five-factor models [10], are grounded in economic intuition and the Arbitrage Pricing Theory (APT) [26]. However, linear assumptions struggle with increasingly complex markets, pronounced nonlinearity [13], and high-dimensional firm characteristics [14].

These challenges have driven substantial interest in data-driven deep learning for asset pricing [7, 11, 15, 24, 39]. Gu et al. [15] show that modern deep learning models can outperform traditional linear baselines when leveraging rich characteristics. Subsequent work improves temporal and cross-sectional modeling, including

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recurrent and attention-based architectures [39], and Transformers and Graph Neural Networks for long-range temporal patterns and cross-asset interactions [7, 11]. These models have been applied to stock return prediction, risk management, and portfolio optimization [12, 28, 36, 40].

Building on this line, deep generative factor models provide a principled way to represent latent market structure under uncertainty. Autoencoder-based methods, particularly the Conditional Autoencoder (CAE) [16], are influential because their encoder-decoder structure aligns naturally with latent factor representations while enabling nonlinear mappings from characteristics to expected returns. VAE-based methods further introduce probabilistic modeling. FactorVAE [8] integrates VAEs into dynamic factor models, and HireVAE [34] and RSAP-DFM [35] incorporate hierarchical and regime-switching mechanisms to address non-stationarity. More recent frameworks such as CAVB [20], Enhancer [3], and Digger-Guider [21] incorporate adversarial learning, meta-learning, and high-frequency factor generation to improve predictive performance under historical shifts. Nevertheless, regime shifts can induce non-Gaussian, multi-modal, and heavy-tailed conditional factor distributions, where standard Gaussian variational parameterizations can become overly smooth and may suffer from posterior collapse [2, 22]. These limitations motivate our use of conditional diffusion [17, 30] as a market-level factor generator. By learning conditional distributions through iterative denoising, diffusion offers a more expressive alternative for complex market stochasticity and extreme dynamics, and has shown promise in financial modeling [18]. For example, DHMoE [4] leverages diffusion to generate a hierarchical mixture-of-experts that captures both short-term fluctuations and long-term trends in a multi-granular manner.

Stronger factor generation alone is not sufficient for robustness. Financial markets experience frequent distribution shifts, so correlations learned in one regime can become spurious in another, leading to severe out-of-distribution degradation. Cochrane [6] highlights the "factor zoo", noting that many proposed factors exhibit weak or unstable performance across periods, illustrating that simply adding more factors does not guarantee robustness. Many adaptive and regime-switching methods still prioritize regime-specific accuracy rather than identifying mechanisms that remain invariant across environments [1]. From a causality-inspired representation learning perspective, robust generalization requires representations that capture stable return-relevant factors rather than regime-specific correlations. While the distribution of characteristics may vary across regimes, some return-relevant structure may remain more stable than regime-specific correlations, consistent with invariant prediction and causality-inspired representation learning [1, 37].

Motivated by this view, we propose the **Causal-Invariant Dynamic Factor Model (Causal-DFM)**. Causal-DFM jointly improves both components of a dynamic factor model by combining diffusion-based factor generation with an Information Bottleneck objective that explicitly regularizes the learned market state. The diffusion component models systematic market factors under rapidly changing regimes through iterative refinement [17, 30], providing a flexible generative mechanism beyond standard variational factor generators [4, 18]. Meanwhile, the market state encoder is trained under the Information Bottleneck principle [31], implemented through an auxiliary prediction proxy, so that the representation remains compact yet strongly return-predictive while suppressing environment-specific variations. This bottlenecked state then parameterizes time-varying factor loadings, encouraging both factor generation and cross-sectional pricing to rely on a shared compact and stable representation rather than regime-specific correlations. By promoting stability via Information Bottleneck regularization while modeling rich factor dynamics generatively, Causal-DFM improves generalization to previously unseen market environments.

In summary, this paper makes the following contributions:

- We formulate stock return prediction from the causality-inspired stable representation perspective, targeting representations that remain stable across heterogeneous market environments.
- We propose Causal-DFM, which improves dynamic factor modeling by using diffusion-based factor generation and an Information Bottleneck regularizer implemented through an auxiliary prediction proxy to learn compact and stable representations for factor loadings, extending prior VAE-based generative factor models.
- We provide theory and experiments supporting improved stable prediction and superior out-of-distribution performance under distribution shifts.

2 Related Work

2.1 Deep Learning for Stock Return Prediction

Stock return prediction has increasingly shifted from traditional statistical modeling to deep learning, motivated by the need to capture nonlinear structure in high-dimensional financial data. Early studies model temporal dependencies using recurrent networks and attention mechanisms [39], and later work incorporates Transformers and Graph Neural Networks to represent long-range temporal dynamics and cross-sectional interactions [7, 11]. Beyond financial prediction, recent spatial-temporal learning studies also show that integrating sequential dynamics with structural knowledge can improve representation quality in dynamic decision tasks [42, 43]. With representation learning, autoencoder-based asset pricing has become a dominant paradigm, where the Conditional Autoencoder (CAE) [16] demonstrates that nonlinear mappings from firm characteristics to expected returns can substantially improve cross-sectional prediction. Variational Autoencoders further introduce probabilistic latent variable modeling to address the low signal-to-noise ratio in financial markets, exemplified by FactorVAE [8] for uncertainty-aware factor generation in dynamic factor frameworks. To handle non-stationarity, HireVAE [34] leverages hierarchical latent structures and online updating under regime changes, while RSAP-DFM [35] introduces explicit regime-switching and posterior factor adaptation. More recent methods such as CAVB [20] and Enhancer [3] combine variational inference with adversarial learning, temporal relational modeling, and meta-learning to improve responsiveness to distribution shifts. Despite architectural differences, these approaches largely optimize predictive accuracy within observed environments by exploiting historical statistical regularities, which can lead to sensitivity to environment-specific

correlations and degraded out-of-distribution performance when regimes change.

2.2 Causal Representation Learning for Robust Forecasting

Causal representation learning seeks latent variables that capture invariant mechanisms underlying observations, thereby improving robustness under distribution shifts. Invariant Risk Minimization [1] formalizes this goal by learning representations for which the optimal predictor remains invariant across environments, and subsequent work further develops structural and information-theoretic perspectives to separate minimal sufficient causes from spurious correlations [37]. Recent advances demonstrate the value of causal invariance in spatio-temporal settings, including TrajCL [23] for disentangling environmental interference in trajectory modeling and CausalDiffRec [41] for mitigating environment-induced confounding in recommender systems through causal diffusion processes. In financial markets, however, causal representation learning remains relatively underexplored, as most stock prediction methods emphasize adaptability to historical regime changes rather than invariance, even though adaptability does not preclude reliance on environment-specific correlations. Building on this gap, our work draws inspiration from causal representation learning and connects it with deep generative factor modeling by using an Information Bottleneck objective [31] to suppress environment-dependent variations while preserving predictive information shared across environments, aligning the modeling objective with causality-inspired stable representation principles for robust forecasting.

3 Preliminary

3.1 Problem Formulation

We study stock return prediction from the structural causality-inspired representation learning perspective. Let $\mathcal{U} = \{1, \ldots, N\}$ denote the stock universe and $\mathcal{T} = \{1, \ldots, T\}$ denote the timeline. For each stock $i \in \mathcal{U}$ at time $t \in \mathcal{T}$, we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^D$ constructed from multi-source information, including price-volume history and firm fundamentals. The prediction target is the next-step excess return $R_{i,t+1} \in \mathbb{R}$.

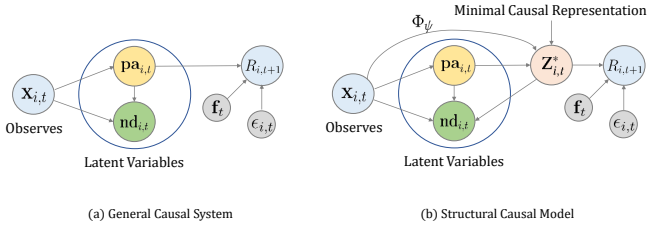
A key challenge is that $\mathbf{x}_{i,t}$ entangles stable return-relevant drivers with environment-dependent variations. Models optimized only for in-sample accuracy may exploit correlations specific to the training regimes and degrade under market distribution shifts. We therefore aim to learn a compact representation $\mathbf{Z}^*_{i,t}$ that preserves predictive information about $R_{i,t+1}$ while suppressing environment-specific dependencies in $\mathbf{x}_{i,t}$. This objective will be instantiated in Section 4.2.

To capture temporal dependencies, we construct a sequential input tensor $\mathbf{S}_{i,t} \in \mathbb{R}^{L \times D}$ by stacking features over a lookback window of length L:

$$\mathbf{S}_{i,t} = \left[\mathbf{x}_{i,t-L+1}, \ldots, \mathbf{x}_{i,t}\right]^\top \in \mathbb{R}^{L \times D}. \tag{1}$$

3.2 Structural Causal Model

To distinguish stable return-relevant signal from spurious variations, we use a Structural Causal Model (SCM) as a motivating assumption, as illustrated in Figure 1.



(a) General Causal System (b) Structural Causal Model

Figure 1: Structural Causal Model for Stock Returns.

Definition 3.1 (SCM for Stock Returns). At each time step t, the data are generated by a causal model $\mathcal{M} = \langle \mathbf{pa}, \mathbf{nd}, \mathbf{f}, R, \mathbf{X} \rangle$, where $\mathbf{X}_{i,t}$ denotes the observable feature random variable (with realization $\mathbf{x}_{i,t}$), $\mathbf{f}_t \in \mathbb{R}^{d_f}$ denotes systematic market factors shared across assets, $\mathbf{pa}_{i,t} \in \mathbb{R}^{d_{pa}}$ denotes asset-specific stable return-relevant drivers that determine exposures to $\mathbf{f}_t$, and $\mathbf{nd}_{i,t} \in \mathbb{R}^{d_{nd}}$ denotes environment-dependent confounders that may correlate with returns within particular regimes but do not follow a stable causal relationship. The observable features are generated by a deterministic nonlinear mixing function h:

$$\mathbf{x}_{i,t} = h(\mathbf{pa}_{i,t}, \mathbf{nd}_{i,t}), \tag{2}$$

and the future return follows an invariant mechanism:

$$R_{i,t+1} = g(\mathbf{pa}_{i,t})^\top \mathbf{f}_t + \varepsilon_{i,t}, \tag{3}$$

where $g(\cdot) : \mathbb{R}^{d_{pa}} \to \mathbb{R}^{d_f}$ maps causal drivers to the d_f-dimensional factor loading vector, and $\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma^2)$ is idiosyncratic noise independent of $\mathbf{pa}_{i,t}$ and $\mathbf{f}_t$.

Eq. (2) formalizes that the observed features conflate invariant causal drivers and regime-dependent confounders through a nonlinear mixing, whereas Eq. (3) specifies that returns depend on asset states only through the causal drivers, via the factor loading function $g(\mathbf{pa}_{i,t})$, together with market-wide factors $\mathbf{f}_t$. Therefore, the shared role of $\mathbf{pa}_{i,t}$ in both equations highlights the core challenge: $\mathbf{pa}_{i,t}$ is not directly observed, yet it is the stable component that governs return predictability across environments.

Our goal is to learn an encoder $\Phi_\psi : \mathbb{R}^{L \times D} \to \mathbb{R}^{d_z}$ that extracts a compact representation $\mathbf{Z}^*_{i,t}$ from the input sequence $\mathbf{S}_{i,t}$, such that $\mathbf{Z}^*_{i,t}$ is encouraged to preserve information related to $\mathbf{pa}_{i,t}$ while suppressing environment-dependent variations. In Figure 1(b), $\mathbf{Z}^*_{i,t}$ serves as a minimal causal representation inferred from observations; in our model, it is further used to parameterize time-varying factor loadings as a surrogate for the unknown $g(\mathbf{pa}_{i,t})$.

The SCM relies on three assumptions:

- *Invariance of return mechanism.* The conditional distribution $P(R_{i,t+1} \mid \mathbf{pa}_{i,t}, \mathbf{f}_t)$ remains stable across market regimes.
- *Environment-dependent confounding.* The marginal distribution $P(\mathbf{nd}_{i,t})$ and its induced statistical dependence in the observations vary across environments, reflecting regime-dependent trading behavior and market microstructure.
- *Entanglement in observations.* Observable features $\mathbf{x}_{i,t}$ conflate causal drivers $\mathbf{pa}_{i,t}$ and confounders $\mathbf{nd}_{i,t}$ through the nonlinear mixing function in Eq. (2).

4 Methodology

In this section, we present the **Causal-Invariant Dynamic Factor Model (Causal-DFM)**. Following the SCM motivation in Figure 1(b), Causal-DFM predicts returns using latent market factors together with asset-specific loadings inferred from a learned compact stable representation $\mathbf{Z}^*_{i,t}$ that is encouraged to capture stable return-relevant information. We model asset returns using the linear factor pricing framework grounded in Arbitrage Pricing Theory (APT) and empirical asset pricing studies [9, 10, 26], and adapt it for predictive tasks as employed in recent generative approaches such as CQVAE [38]:

$$\hat{R}_{i,t+1} = \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_t, \tag{4}$$

where $\mathbf{f}_t \in \mathbb{R}^{d_f}$ denotes the market factor vector generated by a conditional diffusion model (Section 4.3), and $\boldsymbol{\beta}_{i,t} \in \mathbb{R}^{d_f}$ denotes the factor loading vector predicted from the causal representation $\mathbf{Z}^*_{i,t}$ under an information bottleneck regularizer (Section 4.2). The factor dimension d_f is a hyperparameter.

Causal-DFM consists of four components: (1) data preprocessing, (2) causal representation learning with an information bottleneck, (3) diffusion-based factor generation, and (4) factor pricing and end-to-end optimization. The detailed model diagram is shown in Figure 2.

4.1 Data Preprocessing

We preprocess inputs to mitigate scale heterogeneity and improve conditioning under non-stationarity.

4.1.1 Cross-Sectional Normalization. At each time step t, for each feature dimension $d \in \{1, \dots, D\}$ across all N stocks, we compute cross-sectional standardization:

$$\hat{x}_{i,t}^{(d)} = \frac{x_{i,t}^{(d)} - \mu_t^{(d)}}{\sigma_t^{(d)} + \xi}, \tag{5}$$

where $\mu_t^{(d)} = \frac{1}{N}\sum_{j=1}^{N} x_{j,t}^{(d)}$ and $\sigma_t^{(d)} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(x_{j,t}^{(d)} - \mu_t^{(d)}\right)^2}$. The constant $\xi > 0$ is used for numerical stability. This transformation reduces contemporaneous market-wide shifts shared across assets and emphasizes cross-sectional differentiation.

4.1.2 Sliding Window Construction. For each stock i at time t, we construct the normalized input sequence $\hat{\mathbf{S}}_{i,t} \in \mathbb{R}^{L\times D}$ by stacking the past L observations:

$$\hat{\mathbf{S}}_{i,t} = \left[\hat{\mathbf{x}}_{i,t-L+1}, \dots, \hat{\mathbf{x}}_{i,t}\right]^\top. \tag{6}$$

4.2 Information Bottleneck for Causal Representation

The factor generator captures market-wide dynamics (detailed in Section 4.3), but robust prediction further requires asset representations that are less sensitive to regime-specific correlations. We adopt an information bottleneck perspective [31] and the causality-inspired stability motivation in CaRI [37] by learning a compact representation that remains predictive while reducing dependence on environment-dependent variations.

4.2.1 Encoder: Temporal and Cross-Sectional Transformer. Given the normalized sequence input $\hat{\mathbf{S}}_{i,t} \in \mathbb{R}^{L\times D}$, we extract an asset representation using a dual-axis Transformer encoder. We first embed the input by linear projection and positional encoding:

$$\mathbf{H}_{i,t}^{(0)} = \mathrm{PE}\left(\hat{\mathbf{S}}_{i,t}\mathbf{W}_p + \mathbf{b}_p\right) \in \mathbb{R}^{L\times d_z}, \tag{7}$$

where $\mathbf{W}_p \in \mathbb{R}^{D\times d_z}$ and $\mathbf{b}_p \in \mathbb{R}^{d_z}$ are learnable parameters, and $\mathrm{PE}(\cdot)$ denotes sinusoidal positional encoding.

Temporal encoding. To prevent future information leakage, we use causal self-attention:

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top + \mathbf{M}}{\sqrt{d_k}}\right)\mathbf{V}, \tag{8}$$

where $\mathbf{M}$ is a causal mask with $M_{ij} = -\infty$ for $j > i$ and $M_{ij} = 0$ otherwise. Stacking L_{time} causal Transformer layers yields

$$\mathbf{H}_{i,t}^{\text{time}} = \mathcal{T}_\psi^{\text{time}}\left(\mathbf{H}_{i,t}^{(0)}\right) \in \mathbb{R}^{L\times d_z}. \tag{9}$$

Cross-sectional encoding. For each time index $\phi \in \{1, \dots, L\}$, we collect cross-sectional hidden states across stocks:

$$\mathbf{H}_\phi^{\text{cs}} = \begin{bmatrix} \mathbf{H}_{1,t}^{\text{time}}[\phi,:] \\ \vdots \\ \mathbf{H}_{N,t}^{\text{time}}[\phi,:] \end{bmatrix} \in \mathbb{R}^{N\times d_z}. \tag{10}$$

We apply L_{cs} Transformer layers independently on each $\mathbf{H}_\phi^{\text{cs}}$:

$$\widetilde{\mathbf{H}}_\phi^{\text{cs}} = \mathcal{T}_\psi^{\text{cs}}\left(\mathbf{H}_\phi^{\text{cs}}\right) \in \mathbb{R}^{N\times d_z}. \tag{11}$$

We take the last time index to obtain the asset representation:

$$\mathbf{Z}^*_{i,t} = \widetilde{\mathbf{H}}_L^{\text{cs}}[i,:] \in \mathbb{R}^{d_z}. \tag{12}$$

4.2.2 Information Bottleneck Regularization. The classical information bottleneck objective involves mutual information terms that are generally intractable [31]. Consistent with our implementation, we adopt a deterministic bottleneck, where capacity is controlled by the representation dimension and regularization in the encoder, and predictivity is encouraged through an auxiliary proxy.

To motivate the auxiliary proxy, we start from the mutual information decomposition:

$$I(\mathbf{Z}^*_{i,t}; R_{i,t+1}) = H(R_{i,t+1}) - H(R_{i,t+1} \mid \mathbf{Z}^*_{i,t}). \tag{13}$$

Under a fixed data distribution, $H(R_{i,t+1})$ does not depend on the encoder parameters, maximizing $I(\mathbf{Z}^*_{i,t}; R_{i,t+1})$ is equivalent to minimizing $H(R_{i,t+1} \mid \mathbf{Z}^*_{i,t})$.

To obtain a tractable surrogate, let $p(R_{i,t+1} \mid \mathbf{Z}^*_{i,t})$ denote the (unknown) true conditional density. For any variational conditional model $q_\eta(R_{i,t+1} \mid \mathbf{Z}^*_{i,t})$, the non-negativity of KL divergence implies

$$\begin{aligned} &\mathrm{KL}\big(p(\cdot \mid \mathbf{Z}^*_{i,t}) \,\|\, q_\eta(\cdot \mid \mathbf{Z}^*_{i,t})\big) \geq 0 \\ &\implies \mathbb{E}\big[-\log p(R_{i,t+1} \mid \mathbf{Z}^*_{i,t})\big] \leq \mathbb{E}\big[-\log q_\eta(R_{i,t+1} \mid \mathbf{Z}^*_{i,t})\big]. \end{aligned} \tag{14}$$

i.e., $H(R_{i,t+1} \mid \mathbf{Z}^*_{i,t})$ is upper bounded by the expected negative log-likelihood under q_η, where the expectation is taken w.r.t. the joint $p(R_{i,t+1}, \mathbf{Z}^*_{i,t})$. Consequently,

$$I(\mathbf{Z}^*_{i,t}; R_{i,t+1}) \geq H(R_{i,t+1}) + \mathbb{E}\big[\log q_\eta(R_{i,t+1} \mid \mathbf{Z}^*_{i,t})\big], \tag{15}$$

so maximizing $I(\mathbf{Z}^*_{i,t}; R_{i,t+1})$ can be implemented by maximizing the expected log-likelihood under q_η (equivalently, minimizing its expected negative log-likelihood).

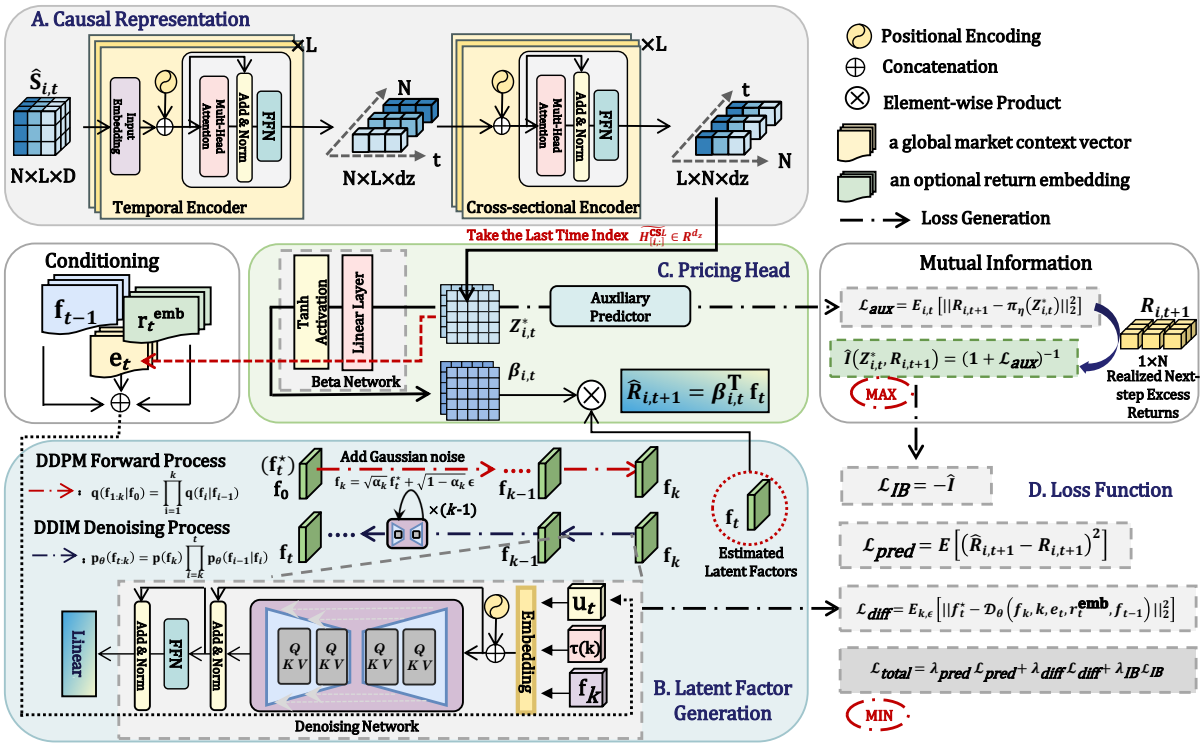


Figure 2: Model Diagram of Causal-DFM.

With real-valued returns, we instantiate q_η as a homoscedastic Gaussian conditional model

$$q_\eta(R_{i,t+1} \mid \mathbf{Z}^*_{i,t}) = \mathcal{N}\big(R_{i,t+1};\, \pi_\eta(\mathbf{Z}^*_{i,t}), \sigma^2\big). \tag{16}$$

Then the negative log-likelihood reduces (up to an additive constant) to a scaled squared error, yielding the auxiliary regression objective below.

We introduce an auxiliary predictor $\pi_\eta(\mathbf{Z}^*_{i,t})$ (implemented as a linear head) to predict the next-step excess return. The auxiliary loss is:

$$\mathcal{L}_{\text{aux}} = \mathbb{E}_{i,t}\left[\left\|R_{i,t+1} - \pi_\eta(\mathbf{Z}^*_{i,t})\right\|_2^2\right]. \tag{17}$$

Following our implementation, we define a stable monotone proxy of mutual information:

$$\widehat{I}\big(\mathbf{Z}^*_{i,t}; R_{i,t+1}\big) = \frac{1}{1+\mathcal{L}_{\text{aux}}}, \tag{18}$$

which increases when $\mathbf{Z}^*_{i,t}$ becomes more predictive of $R_{i,t+1}$. We use the negative proxy as the bottleneck regularizer:

$$\mathcal{L}_{\text{IB}} = -\widehat{I}\big(\mathbf{Z}^*_{i,t}; R_{i,t+1}\big). \tag{19}$$

The choice $\widehat{I} = 1/(1+\mathcal{L}_{\text{aux}})$ is bounded and strictly monotone:

$$0 < \widehat{I}\big(\mathbf{Z}^*_{i,t}; R_{i,t+1}\big) \le 1, \qquad \frac{\partial \widehat{I}}{\partial \mathcal{L}_{\text{aux}}} = -\frac{1}{(1+\mathcal{L}_{\text{aux}})^2} < 0,$$

which prevents regularizer magnitude from becoming unbounded under large auxiliary errors and facilitates balancing with other loss terms. Moreover, if one optimizes $\mathcal{L}_{\text{IB}}$ directly, it corresponds to a gradient-rescaled variant of the auxiliary loss:

$$\frac{\partial \mathcal{L}_{\text{IB}}}{\partial \mathcal{L}_{\text{aux}}} = \frac{1}{(1+\mathcal{L}_{\text{aux}})^2}. \tag{20}$$

In practice, since $\mathcal{L}_{\text{IB}}$ is monotone in $\mathcal{L}_{\text{aux}}$, we optimize the equivalent auxiliary regression loss in Eq. (17) directly for numerical stability.

4.3 Latent Factor Generation via Conditional Diffusion Model

We model market factors with a conditional diffusion generator based on denoising diffusion probabilistic models [17]. The generator produces a time-t market factor vector $\mathbf{f}_t$ that summarizes shared market dynamics and is used in Eq. (4). Diffusion supports iterative refinement during generation and provides a flexible mechanism for representing complex factor behaviors under noisy observations.

4.3.1 Supervision via Pseudo Factor Targets. Training the diffusion generator requires a target to denoise toward. We construct pseudo targets by fitting the cross-sectional pricing relation to realized returns in the training data. Let $\mathbf{R}_{t+1} \in \mathbb{R}^N$ denote the vector of realized next-step excess returns aligned with features at time t, and let $\boldsymbol{\beta}_{i,t}$ be the current factor loadings. We compute the pseudo target factor $\mathbf{f}_t^\star$ by least squares:

$$\mathbf{f}_t^\star = \arg\min_{\mathbf{f}} \sum_{i=1}^{N} \left(R_{i,t+1} - \boldsymbol{\beta}_{i,t}^\top \mathbf{f}\right)^2. \tag{21}$$

The pseudo target is used only to define the denoising objective. In implementation, $\mathbf{f}_t^\star$ is recomputed from the current factor loadings at each training step but is treated as a stop-gradient target when optimizing the denoising loss. The factor used for prediction is generated by the diffusion sampler rather than directly taking $\mathbf{f}_t^\star$. Thus, $\mathbf{f}_t^\star$ serves as a supervised anchor for factor generation instead of a differentiable path that feeds the pricing output back into itself.

4.3.2 Forward Diffusion Process. Given $\mathbf{f}_t^\star$, the forward process progressively corrupts it with Gaussian noise over K_{diff} diffusion steps, following the DDPM framework [17]. At step $k \in \{1, \dots, K_{\text{diff}}\}$, we sample:

$$\mathbf{f}_k = \sqrt{\alpha_k}\,\mathbf{f}_t^\star + \sqrt{1-\alpha_k}\,\boldsymbol{\epsilon}, \tag{22}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\alpha_k = \prod_{j=1}^{k}(1-\omega_j)$ is determined by a predefined noise schedule $\{\omega_j\}$.

4.3.3 Conditional Denoiser. We learn a denoiser that predicts the clean factor from a noisy state while conditioning on market context available at time t. The context combines three signals. First, we summarize the cross-sectional state by mean pooling asset representations to obtain a global market context vector:

$$\mathbf{e}_t = \frac{1}{N}\sum_{i=1}^{N} \mathbf{Z}_{i,t}^* \in \mathbb{R}^{d_z}. \tag{23}$$

Second, we encode contemporaneous return information through a lightweight return encoder and aggregate it by mean pooling, yielding an optional return embedding denoted by $\mathbf{r}_t^{\text{emb}}$. Third, we include the previous-step factor $\mathbf{f}_{t-1}$ to encourage temporal continuity in factor generation.

The denoiser is written as

$$\hat{\mathbf{f}}_t^\star = \mathcal{D}_\theta\Big(\mathbf{f}_k, k, \mathbf{e}_t, \mathbf{r}_t^{\text{emb}}, \mathbf{f}_{t-1}\Big), \tag{24}$$

where denoiser $\mathcal{D}_\theta$ is implemented with a compact Transformer backbone for diffusion denoising, following the diffusion models with Transformer denoisers [25]. Let $\boldsymbol{\tau}(k)$ denote a sinusoidal embedding of the diffusion step. We fuse conditioning signals through a learned mapping:

$$\mathbf{u}_t = C\Big([\mathbf{e}_t; \mathbf{r}_t^{\text{emb}}; \mathbf{f}_{t-1}]\Big), \tag{25}$$

and combine the noisy factor, the fused condition, and the step embedding:

$$\mathbf{h} = \mathbf{W}_{\text{in}}\mathbf{f}_k + \mathbf{u}_t + \mathbf{W}_\tau \boldsymbol{\tau}(k). \tag{26}$$

We apply L_{denoise} Transformer blocks to $\mathbf{h}$ and project the result back to factor space:

$$\hat{\mathbf{f}}_t^\star = \mathbf{W}_{\text{out}} \cdot \mathcal{T}^{(L_{\text{denoise}})} \circ \cdots \circ \mathcal{T}^{(1)}(\mathbf{h}), \tag{27}$$

where each $\mathcal{T}^{(\ell)}$ contains multi-head self-attention and a feed-forward sublayer with residual connections, and $\circ$ denotes function composition, i.e.,

$$\mathcal{T}^{(L_{\text{denoise}})} \circ \cdots \circ \mathcal{T}^{(1)}(\mathbf{h}) = \mathcal{T}^{(L_{\text{denoise}})}(\cdots \mathcal{T}^{(1)}(\mathbf{h}) \cdots).$$

4.3.4 Return-Condition Masking. Return embeddings can improve conditioning during training, but they may be unavailable or undesirable at inference time. To support generation without return inputs, we randomly mask the return embedding during training by setting $\mathbf{r}_t^{\text{emb}} = \mathbf{0}$ with probability p_{mask}. This trains both conditional and unconditioned generation within a single denoiser.

4.3.5 Training Objective. We train a denoiser to predict the pseudo target factor under random noise levels, where gradients are not propagated through the least-squares construction of $\mathbf{f}_t^\star$. We sample diffusion steps uniformly and minimize mean squared error:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{k,\boldsymbol{\epsilon}}\left[\left\|\mathbf{f}_t^\star - \mathcal{D}_\theta\Big(\mathbf{f}_k, k, \mathbf{e}_t, \mathbf{r}_t^{\text{emb}}, \mathbf{f}_{t-1}\Big)\right\|_2^2\right], \tag{28}$$

where $\mathbf{f}_k$ is constructed by Eq. (22).

4.3.6 Inference with DDIM Sampling. At inference time, we generate factors with DDIM sampling [30] for efficiency. DDIM enables deterministic sampling with significantly fewer steps compared to standard DDPM [17], reducing computational cost while maintaining generation quality. We start from Gaussian noise and iteratively denoise conditioned on $\mathbf{e}_t$ and $\mathbf{f}_{t-1}$, optionally using $\mathbf{r}_t^{\text{emb}}$ if available. The final sample is taken as the market factor $\mathbf{f}_t$ and is used in Eq. (4).

4.4 Asset Pricing Model

We integrate the learned representation into a factor pricing head. Given $\mathbf{Z}_{i,t}^*$, we predict the factor loading vector through a Beta network:

$$\boldsymbol{\beta}_{i,t} = \tanh\big(\mathcal{B}_\gamma(\mathbf{Z}_{i,t}^*)\big) \in [-1,1]^{d_f}, \tag{29}$$

where $\mathcal{B}_\gamma$ is a two-layer feedforward network with LayerNorm and GELU activation, matching the implementation in our prediction head. The tanh nonlinearity bounds the loadings and improves numerical stability by preventing excessively large sensitivities.

Given the market factor $\mathbf{f}_t$ generated by Section 4.3, we predict the next-step excess return by Eq. (4).

4.5 Training Objective and Optimization

We jointly optimize the factor generator, the representation encoder, and the pricing head with a weighted sum of three terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{pred}}\mathcal{L}_{\text{pred}} + \lambda_{\text{diff}}\mathcal{L}_{\text{diff}} + \lambda_{\text{IB}}\mathcal{L}_{\text{IB}}, \tag{30}$$

where $\mathcal{L}_{\text{pred}}$ is mean squared error between predicted and realized returns:

$$\mathcal{L}_{\text{pred}} = \mathbb{E}\left[\left(\hat{R}_{i,t+1} - R_{i,t+1}\right)^2\right], \tag{31}$$

$\mathcal{L}_{\text{diff}}$ is the diffusion denoising loss in Eq. (28), and $\mathcal{L}_{\text{IB}}$ is the bottleneck regularizer defined in Eq. (19). This joint objective aligns factor generation and representation learning with the downstream pricing task while discouraging reliance on environment-dependent correlations.

5 Experiments

In this section, we conduct comprehensive experiments to evaluate the performance of the Causal-DFM model. Specifically, we seek to address the following research questions (RQ).

- RQ1 Does our method outperform the state-of-the-art methods in terms of predictive accuracy and maintain a reasonable effectiveness-efficiency trade-off?
- RQ2 Is the improvement in effectiveness mainly due to the proposed diffusion-based framework and the causality-inspired stable learning components?
- RQ3 Can our model's superior prediction performance generate superior investment performance?

Table 1: Asset return prediction performance for different methods on CSI 300, ZZ 500, and NASDAQ datasets. Time denotes the average training time per epoch in seconds on NASDAQ. Best predictive values are highlighted in bold.

Model	CSI300					ZZ500					NASDAQ					
	IC	ICIR	Rank IC	R_{total}	R_{predict}	IC	ICIR	Rank IC	R_{total}	R_{predict}	IC	ICIR	Rank IC	R_{total}	R_{predict}	Time
GRU	0.000 (0.056)	0.001	0.006 (0.075)	-2.078 (0.945)	-2.037 (0.971)	0.001 (0.057)	0.002	0.007 (0.076)	-2.075 (0.948)	-2.034 (0.974)	-0.035 (0.255)	-0.136	-0.043 (0.217)	-1.792 (1.017)	-1.214 (1.080)	1.85
MLP	-0.010 (0.141)	-0.069	-0.003 (0.117)	-0.243 (0.221)	-0.237 (0.222)	-0.009 (0.142)	-0.068	-0.002 (0.118)	-0.240 (0.223)	-0.234 (0.224)	-0.004 (0.203)	-0.020	-0.011 (0.205)	-0.659 (0.892)	-0.679 (0.877)	1.71
Transformer	-0.021 (0.175)	-0.121	-0.016 (0.178)	-0.002 (0.002)	-0.002 (0.002)	-0.020 (0.176)	-0.120	-0.015 (0.179)	-0.001 (0.003)	-0.001 (0.003)	-0.020 (0.202)	-0.100	-0.033 (0.203)	-0.496 (0.006)	-0.484 (0.006)	6.17
CAE	-0.035 (0.207)	-0.174	-0.042 (0.200)	-0.002 (0.003)	-0.002 (0.003)	-0.032 (0.208)	-0.173	-0.041 (0.201)	-0.001 (0.004)	-0.001 (0.004)	0.052 (0.298)	0.174	0.036 (0.299)	-0.004 (0.012)	-0.005 (0.008)	4.47
DY-GAP	0.019 (0.093)	**0.208**	0.010 (0.085)	-0.326 (1.177)	-0.226 (0.236)	0.020 (0.094)	0.176	0.011 (0.086)	-0.323 (1.179)	-0.223 (0.237)	-0.006 (0.182)	-0.035	-0.011 (0.176)	-0.007 (0.830)	-0.007 (0.835)	3.74
MASTER	-0.001 (0.180)	-0.007	-0.014 (0.182)	-0.002 (0.003)	-0.002 (0.003)	-0.002 (0.181)	-0.008	-0.013 (0.183)	-0.001 (0.004)	-0.001 (0.004)	0.083 (0.141)	**0.590**	0.063 (0.157)	-0.004 (0.006)	-0.005 (0.006)	7.64
FactorVAE	0.006 (0.220)	0.027	-0.040 (0.243)	-0.003 (0.028)	-0.000 (0.018)	0.007 (0.221)	0.028	-0.039 (0.244)	-0.002 (0.029)	0.001 (0.019)	0.124 (0.405)	0.289	0.197 (0.336)	-0.022 (0.073)	-0.015 (0.067)	5.80
Causal-DFM	**-0.035** (0.175)	-0.197	**-0.052** (0.183)	**-0.001** (0.294)	**-0.000** (0.022)	**-0.033** (0.176)	**-0.196**	**-0.051** (0.184)	**-0.000** (0.295)	**0.001** (0.023)	**0.178** (0.430)	0.415	**0.401** (0.234)	**0.011** (0.642)	**0.010** (0.665)	17.68

Table 2: Ablation study results of Causal-DFM on CSI 300 and ZZ 500 indices. The table shows the performance of different variants by removing key components. Best values are highlighted in bold.

Variant	CSI300					ZZ500				
	IC	ICIR	Rank IC	R_{total}	R_{predict}	IC	ICIR	Rank IC	R_{total}	R_{predict}
w/o Diffusion	0.01	0.05	-0.04	0.00	0.00	0.03	0.06	-0.03	0.01	0.00
w/o BetaNetwork	-0.03	-0.16	-0.05	-0.00	-0.00	-0.03	-0.15	-0.04	-0.00	-0.00
w/o Transformer Encoder	-0.03	-0.14	-0.06	-0.03	-0.01	-0.03	-0.13	-0.05	-0.02	-0.01
w/o IB Loss	-0.03	-0.18	-0.05	-0.00	-0.00	-0.03	-0.17	-0.04	-0.00	0.00
w/o Prediction Loss	-0.03	-0.18	-0.05	-0.00	-0.00	-0.03	-0.17	-0.04	-0.00	0.00
w/o Classifier-Free Guidance	-0.03	-0.19	-0.05	-0.00	-0.00	-0.03	-0.18	-0.04	-0.00	0.00
w/o Denoiser	-0.03	-0.17	-0.05	-0.00	-0.00	-0.03	-0.16	-0.04	-0.00	-0.00
Full Model	**-0.03**	**-0.20**	**-0.05**	**-0.00**	**-0.00**	**-0.03**	**-0.20**	**-0.05**	**-0.00**	**0.00**

5.1 Experiment Settings

We evaluate the models on the China A-share CSI 300 index and the CSI ZZ 500 index. The datasets are obtained from the Tonglian Data Platform[1], covering daily data of 312 asset factors for all stocks in the respective indices from October 10, 2020, to October 20, 2025. To further examine cross-market generalization, we additionally evaluate the models on the US equity market using a dataset constructed from the NASDAQ Composite Index. The US dataset is obtained from the Wharton Research Data Services (WRDS) platform[2], covering monthly data of 107 asset factors for stocks in the index universe from January 31, 1990, to December 31, 2021. The factors span multiple categories including quality, value, growth, momentum, risk, sentiment, technical, and size factors. All factors are standardized to have zero mean and unit variance. Missing values are handled via interpolation, and outliers are treated using winsorization. Each dataset is split chronologically into training (70%), validation (10%), and testing (20%) sets to avoid look-ahead bias. The experiment is conducted with sliding windows of length $L = 20$. Unless otherwise specified, Causal-DFM uses $\lambda_{\text{pred}} = 8.0$, $\lambda_{\text{diff}} = 1.0$, $\lambda_{\text{IB}} = 5 \times 10^{-5}$, $K_{\text{diff}} = 1000$, 30 DDIM sampling steps, return-condition masking probability $p_{\text{mask}} = 0.05$, factor dimension $d_f = 24$, and encoder depth 3.

Causal-DFM is compared against 7 baselines, including both traditional machine learning and deep learning-based approaches. The baselines include: ① GRU [5]; ② MLP [27]; ③ Transformer [33]; ④ CAE [16]; ⑤ DY-GAP [32]; ⑥ MASTER [19]; and ⑦ FactorVAE [8].

The performance is evaluated using five key metrics: Information Coefficient (IC), Information Coefficient Information Ratio (ICIR), Rank Information Coefficient (Rank IC), Total R-squared (R_{total}) and Predictive R-squared (R_{predict}). Formal definitions are provided in the Appendix B. For IC, ICIR, and Rank IC, higher absolute values indicate better performance, while for R_{total} and R_{predict}, higher values are better. For the NASDAQ dataset, we additionally report the average training time per epoch in seconds to evaluate computational efficiency. The metric implementations follow the standardized approaches in [8] and [20].

[1] https://uqer.datayes.com/
[2] https://wrds-web.wharton.upenn.edu/

Table 3: Sensitivity analysis of the information bottleneck loss weight λ_{IB} on the CSI 300 index.

λ_{IB}	IC	ICIR	Rank IC
1×10^{-6}	-0.028	-0.188	-0.046
1×10^{-5}	-0.029	-0.198	-0.046
5×10^{-5}	-0.035	-0.197	-0.052
1×10^{-4}	-0.032	-0.191	-0.052
5×10^{-4}	-0.028	-0.193	-0.049
1×10^{-3}	-0.028	-0.189	-0.051

Table 4: Performance comparison of latent factor generation strategies on the CSI 300 index. Best values are highlighted in bold.

Model	IC	ICIR	Rank IC	R_{total}	R_{predict}
Det.	-0.027	-0.150	-0.048	-0.016	-0.002
VAE	-0.027	-0.150	-0.047	-0.098	-0.017
Diffusion	**-0.035**	**-0.197**	**-0.052**	**-0.001**	**-0.000**

5.2 Prediction Performance and Efficiency (RQ1)

Table 1 summarizes the asset return prediction performance of all models on the test dataset for the CSI 300, ZZ 500, and NASDAQ datasets. For NASDAQ, Table 1 also reports the average training time per epoch to evaluate computational efficiency. Our proposed Causal-DFM model achieves the best performance on the majority of key evaluation metrics across the three datasets, outperforming all baseline methods. On the CSI 300 index, Causal-DFM attains the highest absolute IC value of 0.035, demonstrating its superior forecasting capability. It also yields the lowest (i.e., most negative) Rank IC of -0.052, indicating a strong ability to rank future returns accurately. In terms of R_{total} and R_{predict}, Causal-DFM achieves the best (highest) values of -0.001 and -0.000, respectively. The DY-GAP model achieves the highest ICIR of 0.208 on the CSI 300 index. On the ZZ 500 index, Causal-DFM attains the best performance across all five metrics, achieving the highest absolute IC (0.033), the highest absolute ICIR (0.196), the lowest Rank IC (-0.051), and the highest R_{total} and R_{predict} values (-0.000 and 0.001, respectively). The results confirm that the proposed Causal-DFM model consistently provides the most accurate predictions and the most stable ranking performance across different market indices, validating the effectiveness of its causality-inspired stable learning framework. The NASDAQ results further demonstrate the cross-market effectiveness of Causal-DFM. As shown in Table 1, Causal-DFM achieves the best results on four of the five predictive metrics, including IC, Rank IC, R_{total}, and R_{predict}, while remaining competitive on ICIR. Its average training time is 17.68 seconds per epoch, which is higher than simpler baselines but remains acceptable for stock return prediction scenarios that do not require high-frequency inference. These results suggest a reasonable effectiveness-efficiency trade-off rather than a purely efficiency-oriented advantage.

5.3 Ablation Study and Sensitivity Analysis (RQ2)

To validate the effectiveness of key components in Causal-DFM, we conduct component-level ablations and sensitivity analyses. First, we remove individual modules from the full model to examine their respective contributions. The variants include: (1) removing the diffusion module and replacing it with an MLP (w/o Diffusion); (2) removing the beta network that predicts factor loadings from latent representations (w/o BetaNetwork); (3) removing the temporal cross-sectional Transformer encoder (w/o Transformer Encoder); (4) removing the information bottleneck loss applied to the latent representation $\mathbf{Z}^*_{i,t}$ (w/o IB Loss); (5) removing the prediction loss that directly optimizes the forecasting task (w/o Prediction Loss); (6) removing the classifier-free guidance mechanism in the diffusion process (w/o Classifier-Free Guidance); and (7) removing the denoiser module (w/o Denoiser). Table 2 presents the results on both the CSI 300 and ZZ 500 indices. Second, we analyze the sensitivity of Causal-DFM to the information bottleneck loss weight λ_{IB}, as shown in Table 3. Finally, to further evaluate whether diffusion is a more suitable latent factor generator, we compare it with deterministic and VAE-based alternatives under the same main framework, as reported in Table 4.

The component-level ablation results show that removing key modules generally weakens model performance on both indices, indicating that the proposed architecture benefits from the joint design of causal representation learning, latent factor generation, and factor pricing. The beta network and Transformer encoder are important for modeling dynamic stock-factor relationships and extracting cross-sectional temporal representations. The information bottleneck loss helps the latent representation $\mathbf{Z}^*_{i,t}$ retain return-predictive information while reducing sensitivity to spurious regime-specific correlations. The prediction loss directly aligns the learned factors and loadings with the forecasting objective. In addition, the classifier-free guidance mechanism and the denoiser contribute to stable conditional generation in the reverse diffusion process. These results confirm that the effectiveness of Causal-DFM does not come from a single component, but from the coordinated optimization of representation learning, factor generation, and pricing.

We further study the effect of the information bottleneck loss weight λ_{IB}. As shown in Table 3, moderate IB regularization leads to better ranking performance, with $\lambda_{\text{IB}} = 5 \times 10^{-5}$ achieving the best absolute IC and Rank IC. Very small or large weights yield weaker results, suggesting that an appropriate bottleneck strength is important for balancing predictive information preservation and regime-specific noise suppression.

To further isolate the effect of the latent factor generator, we compare the proposed diffusion generator with two alternative designs under the same encoder, beta network, and training protocol: a deterministic generator and a VAE generator. As shown in Table 4, the diffusion generator achieves the strongest performance across all five metrics on the CSI 300 index. Compared with the deterministic and VAE variants, diffusion improves the absolute IC from 0.027 to 0.035 and the absolute Rank IC from 0.048/0.047 to 0.052, while also yielding better R_{total} and R_{predict}. These results indicate that the iterative denoising process provides a more expressive factor

Table 5: Portfolio performance on the CSI 300 index. AR, SR, and MDD denote annualized return, Sharpe ratio, and maximum drawdown, respectively. Best values are highlighted in bold.

Method	AR (↑)	SR (↑)	MDD (↓)
GRU	1.16%	0.1537	8.56%
MLP	-11.84%	-0.7686	11.38%
Transformer	5.53%	0.4226	7.79%
CAE	-3.22%	-0.0940	10.82%
DY-GAP	11.08%	0.6992	9.17%
MASTER	-8.05%	-0.4997	7.83%
FactorVAE	15.81%	0.9301	11.48%
Causal-DFM	**20.89%**	**1.2051**	**7.61%**

Table 6: Transaction-cost sensitivity of the Causal-DFM portfolio on CSI 300. Costs are measured in bps per trade.

Cost (bps)	AR (↑)	SR (↑)	MDD (↓)
0	42.34%	2.06	6.99%
10	28.69%	1.50	7.47%
30	20.89%	1.21	7.61%
50	-4.92%	-0.19	10.26%

prior for regime-varying and non-Gaussian market dynamics than deterministic or VAE alternatives.

5.4 Portfolio Investment (RQ3)

To evaluate the investment performance of Causal-DFM, we construct a daily rebalanced portfolio based on model predictions. We focus on the China A-share CSI 300 index, from which we select 50 stocks from the index constituents ranked by predicted returns $\hat{R}_{i,t+1}$, with equal weight allocation and a turnover constraint limiting position changes to 10 stocks per rebalance. The portfolio is long-only with no leverage, and transaction costs are set at 30 basis points (bps) per trade in the main comparison, with additional transaction-cost sensitivity analysis reported below.

Table 5 reports the portfolio performance on the CSI 300 index. Causal-DFM achieves the highest AR of 20.89% and SR of 1.2051, while maintaining the lowest MDD of 7.61%. Compared with FactorVAE, the second-best baseline, Causal-DFM improves both return and risk-adjusted return and reduces drawdown, indicating that its predictive advantage translates into practical investment gains. Table 6 further reports transaction-cost sensitivity. As expected, higher costs reduce AR and SR, but Causal-DFM remains positive under the default 30 bps setting. The degradation at 50 bps indicates that turnover control is important for practical deployment.

Figure 3 illustrates the cumulative returns of the constructed portfolios over the test period. The portfolio performance curves demonstrate clear differentiation among models throughout the investment horizon. Causal-DFM exhibits the most consistent and superior growth trajectory, achieving a final cumulative return of

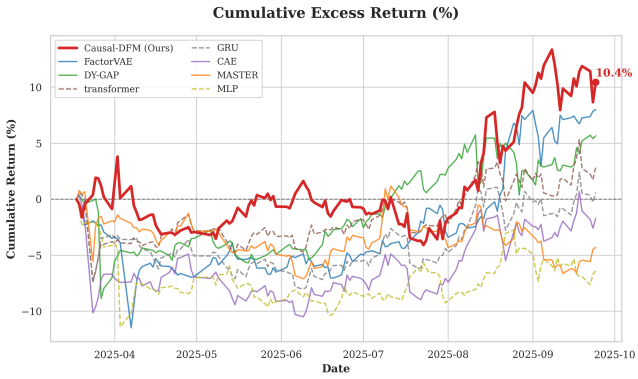


Figure 3: Cumulative portfolio returns on the CSI 300 index.

11.42%, significantly outperforming all baseline models. This performance advantage is particularly evident during market fluctuations, where Causal-DFM maintains stable growth while other models experience more pronounced volatility. The second-best performer, FactorVAE, achieves a cumulative return of 7.21%, while DY-GAP and Transformer show moderate performance with 4.91% and 0.34% respectively. Notably, MLP and CAE demonstrate negative cumulative returns of -8.84% and -7.49%, highlighting the challenges these models face in generating consistent investment value. The sustained outperformance of Causal-DFM throughout the test period validates its robustness and stability in real-world investment applications, effectively translating its superior predictive capabilities into tangible portfolio gains.

6 Conclusion

We propose the Causal-DFM, which combines a conditional diffusion factor prior with an Information-Bottleneck-regularized state encoder to reduce reliance on environment-dependent correlations while preserving a compact, return-predictive representation. We provide an information-theoretic analysis to justify the objective. Experiments on real-world stock datasets show that Causal-DFM consistently outperforms strong baselines, with particularly pronounced gains during regime transitions and crisis periods, demonstrating improved robustness to distribution shift. In future, we will move beyond the limitations of pseudo targets by developing more theoretically grounded training signals for factor generation, extend the framework to handle potentially evolving return mechanisms across regimes, and conduct broader evaluations on additional datasets with more realistic portfolio backtesting protocols.

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A Additional Theoretical Discussion

This appendix provides supporting arguments for the causality-inspired stable representation motivation of Causal-DFM.

A.1 Predictive Sufficiency under the SCM

Proposition A.1 (Predictive sufficiency under the SCM). *Under the SCM in Definition 3.1, assume that environment-dependent confounders carry no additional predictive information about returns*

once the stable return-relevant drivers and market factors are given:

$$I(\mathbf{nd}_{i,t}; R_{i,t+1} \mid \mathbf{pa}_{i,t}, \mathbf{f}_t) = 0. \tag{32}$$

Let $\mathbf{Z}^*_{i,t}$ be any representation computed from $\mathbf{S}_{i,t}$. If $\mathbf{Z}^*_{i,t}$ is predictively sufficient for $R_{i,t+1}$ in the sense that

$$I(\mathbf{Z}^*_{i,t}; R_{i,t+1} \mid \mathbf{f}_t) = I(\mathbf{S}_{i,t}; R_{i,t+1} \mid \mathbf{f}_t), \tag{33}$$

then $\mathbf{Z}^*_{i,t}$ preserves all predictive information carried by the stable return-relevant drivers:

$$I(\mathbf{Z}^*_{i,t}; R_{i,t+1} \mid \mathbf{f}_t) = I(\mathbf{pa}_{i,t}; R_{i,t+1} \mid \mathbf{f}_t). \tag{34}$$

Proof. By the data processing inequality, for any representation $\mathbf{Z}^*_{i,t}$ computed from $\mathbf{S}_{i,t}$,

$$I(\mathbf{Z}^*_{i,t}; R_{i,t+1}) \le I(\mathbf{S}_{i,t}; R_{i,t+1}). \tag{35}$$

By the SCM, $\mathbf{S}_{i,t}$ is a function of $(\mathbf{pa}_{i,t}, \mathbf{nd}_{i,t})$. Using the chain rule,

$$\begin{aligned} I(\mathbf{S}_{i,t}; R_{i,t+1} \mid \mathbf{f}_t) &= I(\mathbf{pa}_{i,t}, \mathbf{nd}_{i,t}; R_{i,t+1} \mid \mathbf{f}_t) \\ &= I(\mathbf{pa}_{i,t}; R_{i,t+1} \mid \mathbf{f}_t) + I(\mathbf{nd}_{i,t}; R_{i,t+1} \mid \mathbf{pa}_{i,t}, \mathbf{f}_t). \end{aligned} \tag{36}$$

Under Eq. (32), the second term vanishes, which yields

$$I(\mathbf{S}_{i,t}; R_{i,t+1}) = I(\mathbf{pa}_{i,t}; R_{i,t+1}).$$

If $\mathbf{Z}^*_{i,t}$ is predictively sufficient, Eq. (33) implies

$$I(\mathbf{Z}^*_{i,t}; R_{i,t+1}) = I(\mathbf{S}_{i,t}; R_{i,t+1}),$$

and therefore Eq. (34) holds. □

A.2 Invariance View of the IB Regularizer

We further clarify why the IB-inspired regularizer is related to cross-environment stable representation learning under the SCM assumptions. This analysis is not intended to prove causal identification. Instead, it shows that, if the SCM assumptions hold, a compact and predictive representation is biased toward retaining stable return-relevant information while discarding environment-dependent nuisance information.

Let E_t denote the market environment at time t, $\mathbf{x}_{i,t}$ the observed feature, $\mathbf{f}_t$ the market factor, $R_{i,t+1}$ the future return, and $\mathbf{Z}^*_{i,t} = \Phi_\psi(\mathbf{x}_{i,t})$ the learned representation. Following Definition 3.1, assume that the observed feature admits the decomposition

$$\mathbf{x}_{i,t} = h(\mathbf{pa}_{i,t}, \mathbf{nd}_{i,t}), \tag{37}$$

where $\mathbf{pa}_{i,t}$ denotes the stable return-relevant component and $\mathbf{nd}_{i,t}$ denotes the environment-dependent nuisance component. We further assume the conditional invariance condition

$$R_{i,t+1} \perp (\mathbf{nd}_{i,t}, E_t) \mid (\mathbf{pa}_{i,t}, \mathbf{f}_t). \tag{38}$$

This condition means that, once the stable causal drivers and market factors are given, the nuisance component and the environment index do not provide additional predictive information about future returns.

First, the compression part of the IB objective suppresses the environment-related information. Since $\mathbf{Z}^*_{i,t}$ is computed from $\mathbf{x}_{i,t}$, the variables satisfy the Markov chain

$$E_t \to \mathbf{x}_{i,t} \to \mathbf{Z}^*_{i,t}. \tag{39}$$

By the data processing inequality,

$$I(E_t; \mathbf{Z}^*_{i,t}) \le I(\mathbf{x}_{i,t}; \mathbf{Z}^*_{i,t}). \tag{40}$$

Therefore, reducing $I(\mathbf{x}_{i,t}; \mathbf{Z}^*_{i,t})$ also reduces an upper bound on the environment information retained in $\mathbf{Z}^*_{i,t}$. This explains why the compression side of IB discourages the representation from encoding regime-specific nuisance variations.

Second, among representations that preserve predictive information, the IB objective prefers a minimal sufficient representation. Consider the constrained form of the IB principle

$$\min_{\Phi_\psi} I(\mathbf{x}_{i,t}; \mathbf{Z}^*_{i,t}) \quad \text{s.t.} \quad I(R_{i,t+1}; \mathbf{Z}^*_{i,t} \mid \mathbf{f}_t) = I(R_{i,t+1}; \mathbf{x}_{i,t} \mid \mathbf{f}_t). \tag{41}$$

The constraint means that $\mathbf{Z}^*_{i,t}$ preserves all predictive information contained in $\mathbf{x}_{i,t}$ once the market factor $\mathbf{f}_t$ is given. Suppose there exists a stable sufficient representation

$$\mathbf{Z}^{\text{inv}}_{i,t} = T(\mathbf{pa}_{i,t}), \tag{42}$$

such that

$$R_{i,t+1} \perp \mathbf{x}_{i,t} \mid (\mathbf{Z}^{\text{inv}}_{i,t}, \mathbf{f}_t). \tag{43}$$

Then $\mathbf{Z}^{\text{inv}}_{i,t}$ is feasible for the constrained problem, because it preserves the return-relevant information in $\mathbf{x}_{i,t}$ through the stable component $\mathbf{pa}_{i,t}$. For any other feasible representation $\mathbf{Z}$, the sufficiency constraint implies

$$R_{i,t+1} \perp \mathbf{x}_{i,t} \mid (\mathbf{Z}, \mathbf{f}_t). \tag{44}$$

Thus, $\mathbf{Z}$ must also be conditionally sufficient for predicting returns. If $\mathbf{Z}^{\text{inv}}_{i,t}$ is the minimal stable sufficient representation, then any feasible $\mathbf{Z}$ must contain at least the information needed to recover $\mathbf{Z}^{\text{inv}}_{i,t}$. Equivalently, $\mathbf{Z}^{\text{inv}}_{i,t}$ can be regarded as a measurable function of $\mathbf{Z}$. Applying the data processing inequality gives

$$I(\mathbf{x}_{i,t}; \mathbf{Z}^{\text{inv}}_{i,t}) \le I(\mathbf{x}_{i,t}; \mathbf{Z}). \tag{45}$$

Therefore, among sufficient predictive representations, the IB objective is biased toward the minimal stable sufficient representation. Since $\mathbf{Z}^{\text{inv}}_{i,t}$ depends only on $\mathbf{pa}_{i,t}$ and $\mathbf{pa}_{i,t}$ is assumed to be stable across environments, $\mathbf{Z}^{\text{inv}}_{i,t}$ is also stable across environments.

Taken together, the above argument establishes the formal sense in which our regularizer is causality-inspired: the compression term suppresses environment-dependent information, while the predictiveness term preserves return-relevant information. Under the SCM assumptions, their combination favors a compact sufficient representation that is aligned with stable return-relevant components. In practice, our implementation uses a deterministic auxiliary-loss surrogate rather than an exact stochastic IB objective, so this result should be interpreted as a theoretical motivation rather than a causal identification guarantee.

A.3 Generalization under Invariant Mechanisms

Theorem A.2 (Generalization under distribution shift). *Let $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ be two distributions that share the same causal mechanism for returns, meaning that the conditional law of R given the true causal drivers and market factors is invariant across distributions. Consider a predictor $\hat{R}(\mathbf{Z}^*, \mathbf{f})$ trained on $\mathcal{D}_{\text{train}}$. Then, with probability at least $1-\delta$, the expected test risk satisfies*

$$\mathbb{E}_{\mathcal{D}_{\text{test}}}[\mathcal{L}] \leq \mathbb{E}_{\mathcal{D}_{\text{test}}}[\mathcal{L}^\star] + O\left(\sqrt{\frac{\log(1/\delta)}{m}}\right) + \epsilon_{\text{approx}}, \tag{46}$$

where $\mathcal{L}^\star$ is the Bayes risk under the invariant mechanism, m is the sample size, and ϵ_{approx} captures the approximation error induced by using $\mathbf{Z}^*$ instead of the true causal drivers. This statement follows the invariance-based analysis framework in [37].

Proof sketch. The proof follows [37]. Under the invariance assumption, the conditional mechanism that determines returns is stable across environments. The test risk can be decomposed into an estimation term due to finite samples, with the standard rate $O(\sqrt{\log(1/\delta)/m})$, and an approximation term induced by replacing the true causal drivers with $\mathbf{Z}^*$. As $\mathbf{Z}^*$ better preserves the predictive information of the causal drivers, the approximation term decreases. The key implication is that robustness is governed by representation quality under an invariant mechanism, rather than by fitting environment-specific correlations in the observed feature space. □

B Evaluation Metrics

Information Coefficient (IC): At each time t, IC is the Pearson correlation between predicted and actual returns across the N stocks:

$$\text{IC}_t = \frac{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{R}_{i,t+1} - \overline{\hat{R}}_{t+1}\right)\left(R_{i,t+1} - \overline{R}_{t+1}\right)}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{R}_{i,t+1} - \overline{\hat{R}}_{t+1}\right)^2}\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(R_{i,t+1} - \overline{R}_{t+1}\right)^2}},$$

where $\overline{\hat{R}}_{t+1} = \frac{1}{N}\sum_{i=1}^{N}\hat{R}_{i,t+1}$ and $\overline{R}_{t+1} = \frac{1}{N}\sum_{i=1}^{N}R_{i,t+1}$. $\hat{R}$ represents predicted returns and R represents actual returns.

Information Coefficient Information Ratio (ICIR): Let $\{\text{IC}_t\}$ be the IC series over the evaluation period, ICIR is defined as:

$$\text{ICIR} = \frac{\mu(\{\text{IC}_t\}_t)}{\sigma(\{\text{IC}_t\}_t)},$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ denote the sample mean and standard deviation over time.

Rank Information Coefficient (Rank IC): At each time t, Rank IC is the Spearman correlation between predicted and actual returns across the N stocks:

$$\tilde{\rho}_{i,t} := \text{rank}(\hat{R}_{i,t+1}) - \overline{\text{rank}(\hat{R})}_{t+1},$$
$$\rho_{i,t} := \text{rank}(R_{i,t+1}) - \overline{\text{rank}(R)}_{t+1},$$
$$\text{RankIC}_t = \frac{\frac{1}{N}\sum_{i=1}^{N}\tilde{\rho}_{i,t}\rho_{i,t}}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\tilde{\rho}_{i,t}^2}\sqrt{\frac{1}{N}\sum_{i=1}^{N}\rho_{i,t}^2}}.$$

where $\overline{\text{rank}(\hat{R})}_{t+1}$ and $\overline{\text{rank}(R)}_{t+1}$ denote the cross-sectional means of $\{\text{rank}(\hat{R}_{i,t+1})\}_{i=1}^{N}$ and $\{\text{rank}(R_{i,t+1})\}_{i=1}^{N}$, respectively.

Total R-squared (R_{total}): Measures the explanatory power using factors reconstructed from current returns:

$$R_{\text{total}} = \left(1 - \frac{\sum_t\sum_{i=1}^{N}(R_{i,t+1} - \hat{R}_{\text{total},i,t+1})^2}{\sum_t\sum_{i=1}^{N}R_{i,t+1}^2}\right) \times 100\%$$

where $\hat{R}_{\text{total}}$ is predicted using current factor representations.

Predictive R-squared (R_{predict}): Evaluates predictive capability using historical factor means:

$$R_{\text{predict}} = \left(1 - \frac{\sum_t\sum_{i=1}^{N}(R_{i,t+1} - \hat{R}_{\text{predict},i,t+1})^2}{\sum_t\sum_{i=1}^{N}R_{i,t+1}^2}\right) \times 100\%$$

where $\hat{R}_{\text{predict}}$ is predicted using averaged historical factor representations.